Notes on *A Hybrid Strategy for Network Immunization*

1 Motivation

1.1 Network Immunization Problem

Definition. Given a network represented by an undirected graph $G = (V, E)$, the *network immunization problem* aims to select an *immunized set* $V_P \subseteq V$ that minimizes the final number of infected nodes V_I.

Objective. Given an immunization budget B, find

$$\min_{V_p \subseteq V} \mathbb{E}[|V_I|],$$
$$\text{subject to } |V_P| \leq B.$$

1.2 Existing Strategies and Hypothesis

Traditional immunization strategies identify important nodes using structural centrality measures, including:

- **Degree.** The number of neighbors
- **Betweenness.** The fraction of shortest paths passing through a node
- **k-core.** How deeply a node is embedded in a densely connected region
- **Eigenvector centrality.** Based on a node's connections to other important nodes

However, topology alone does not determine whether a node is influential in virus propagation. For example, a highly connected but inactive node may contribute less than a less-connected but highly active node. Therefore, Li et al. hypothesizes that combining *structural centrality* with *individual activity* can identify better immunization targets.

2 Methodology

2.1 Modeling Node Activity

Li et al. introduce an activity parameter a_i for each node v_i, representing its participation in virus propagation. To capture heterogeneous behavior in the network, they consider three probability distribution, used to generate different activity patterns across nodes in simulations:

- **Exponential.** Connections between nodes happen at a steady, limited rate due to real-world constraints (e.g. time, energy).
$$f(x; \lambda) = \lambda \cdot \exp(-\lambda x)$$
- **Power law.** Probability of getting a new connection depends on how many connections the node already has (i.e. "the rich get richer").
$$f(x) \propto x^{-\alpha}$$
- **Negative binomial.** Larger variance (i.e. there are distinct subgroups of highly active and highly passive nodes).
$$f(k; r, p) = \binom{k + r - 1}{r - 1} \cdot p^r \cdot (1 - p)^k$$

2.2 Activity-Ranking (AR) Metric

Definition. Li et al. propose an **Activity Ranking** (AR) score that combines structural centrality and activity,

$$AR_i = C_i[-\log(\frac{a_i}{\sum_{j\in V} a_j})^2].$$

- C_i is the centrality (degree, betweenness, k-core, and eigenvector centrality) of node v_i
- a_i is the activity parameter of node v_i
- $\sum_{j\in V} a_j$ is the total activity across the network

Algorithm. (Ranking-based heuristic)

1. Compute each node's structural centrality C_i.
2. Calculate its activity-ranking score AR_i.
3. Rank nodes in descending order of AR_i.
4. Immunize the highest-ranked nodes until the budget is exhausted.

3 Experimental Design

3.1 Propagation Model

Definition. An *email-based propagation model* is an application of the Independent Cascade Model, where each active node u has one chance to influence each inactive neighbor v through an email recommending, succeeding independently with probability p_{uv}.

- The probability that node u successfully activates node v, given that all nodes in X have already tried and failed to activate v is $p_v(u, X) = p_{uv}$.
- Interpretation of the email network:
  - Nodes: Individuals who send/receive email recommendations
  - Edges: $(u, v) \in E \Leftrightarrow u$ can send an email recommendation to v
  - p_{uv}: The probability that v adopts the recommended product after receiving an email from u
  - Active nodes: Individuals who have adopted the product and can recommend it to others

Procedure. For each experiment,

1. Select two randomly initially infected nodes.
2. Randomly divide nodes into three groups, assigning activity characteristics using the three probability distributions.
3. Immunize nodes using either a traditional centrality centric, or its AR counterpart.
4. Simulate propagation and measure the infected proportion over time.
5. Average results over 100 runs.

3.2 Datasets and Baselines

Networks. Li et al. evaluate their method on eight empirical networks (email, scientific coauthorship, and other social networks) and one synthetic community-based network.

Immunization budgets. They test $B_1 = 0.1 \cdot |V|$ and $B_2 = 0.3 \cdot |V|$.

Evaluation metric. They average the proportion of infected nodes at time step t,

$$I(t) = \frac{|V_I(t)|}{|V|},$$

where a lower infection proportion indicate more effective immunization.

4 Results and Limitations

4.1 Experimental Findings

$B_1 = 0.1 \cdot |V|$. (Figure 4)
AR strategies show relatively small improvements over traditional strategies, suggesting that structural centrality dominates when only a small number of nodes can be immunized.

$B_1 = 0.3 \cdot |V|$. (Figure 5)
AR strategies produce more substantial improvements in several networks. Li et al. argues that as more nodes are selected, activity becomes more useful for distinguishing nodes with similar centrality values.

4.2 Limitations

1. There is no consideration of redundancy, as nodes are ranked independently. For example, two highly-ranked nodes may protect overlapping regions of the network, making immunizing both less effective than selecting nodes with complementary coverage.

2. Since activity patterns are assigned using probability distributions rather than observed temporal behavior from each network, real network may instead correlate with factors such as the node's degree, community membership, or its neighbors' behavior.

3. The algorithm selects immunized nodes before virus propagation, so it does not adapt to changes in activity or infection states during the cascade.

4. This approach still depends on the same centrality metrics as traditional strategies, some of which are expensive to compute for large or dynamic networks.

5. Though the authors provide empirical evidence, they do not prove the optimality or any approximation guarantees for the selected immunization set.